Rishabh Dora

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Research Interests

Nonequilibrium quantum dynamics; light–matter interactions; quantum transport; computational condensed matter physics; electronic structure; many-body and open quantum systems; scientific computing and algorithm development.

Education

University of Rochester Sept 2022 – Present
PhD, Chemistry *Rochester, NY*
MS, Chemistry, 2024 (GPA: 3.95/4.0)

Indian Institute of Science Education and Research (IISER) Bhopal Aug 2017 – May 2022
BS-MS Dual Degree, Physics (CPI: 8.64/10.0) *Bhopal, India*

Research Experience

Graduate Research Assistant, University of Rochester 2022 – Present
Quantum Dynamics Simulations, Light–Matter Interactions and Quantum Transport *Advisor: Prof. Ignacio Franco*

Light–Matter Interactions Beyond the Dipole Approximation

- Developed a general theoretical and computational framework for describing light–matter interactions in extended systems under arbitrarily structured electromagnetic fields, going beyond the electric-dipole approximation without truncation at finite multipolar order.
- Integrated spatially resolved electromagnetic fields with Wannier-based, first-principles electronic-structure models and time-dependent quantum dynamics, enabling simulations of realistic nanoscale electromagnetic environments.
- Established the physical regimes in which the dipole approximation remains reliable, clarifying when the spatial structure of light must be retained for predictive simulations of nanoscale devices, interfaces, and quantum materials.

Nonequilibrium Quantum Transport and Light-Driven Charge Control

- Developed a time-dependent nonequilibrium Green's-function (NEGF) framework to investigate ultrafast light-driven transport and optical control of charge carriers in nanoscale junctions.
- Established how laser waveform, phase, and pulse duration govern the generation and separation of light-driven charge carriers, and examined the robustness of this control across graphene nanojunctions and systems spanning metallic, trivial, and topological regimes.
- Parallelized computational bottlenecks using MPI and optimized numerical propagation, reducing runtime of a major simulation workflow by 75%, increasing accessible system sizes, and enabling simulations over longer time scales.

Quantum Transport in Realistic Nanoscale Environments

- Unified the beyond-dipole light–matter framework with time-dependent quantum transport to move from idealized spatially uniform models toward simulations of metal–graphene–metal junctions under experimentally realistic, spatially nonuniform optical fields.
- Investigating how spatially structured electromagnetic fields, bulk graphene, and metal–graphene interfaces jointly determine light-driven transport at petahertz timescales.

Research Assistant, IISER Bhopal 2021 – 2022
Many-Body Dynamics and Open-Quantum Systems *Advisor: Prof. Sebastian Wüster*

- Developed many-body dynamics for coupled photon–phonon systems in optomechanical arrays to investigate polaron formation, localization, tunneling, and quantum transport.
- Applied time-dependent quantum dynamics and polaron transformations to connect microscopic photon–phonon interactions with emergent transport behavior, extending simulations from small exact systems to larger arrays.
- Investigated how coupling strength, oscillator frequency, inter-cavity tunneling, and environmental decoherence determine transport timescales and the robustness of polaron dynamics in engineered quantum systems.

Computational and Research Methods

Quantum Dynamics and Transport: nonequilibrium Green's functions (NEGF), time-dependent quantum dynamics, open quantum systems, Lindblad dynamics, NMQSD, tight-binding models
Electronic Structure and Materials: Wannier90, maximally localized Wannier functions (MLWFs), Quantum ESPRESSO, first-principles/*ab initio* workflows
Machine Learning and Data Analysis: Scikit-Learn, PyTorch; graduate-level machine-learning coursework
Electromagnetic Simulation: MEEP, finite-difference time-domain (FDTD) methods, spatially resolved electromagnetic-field simulations
Scientific Computing and HPC: Python, Fortran, Julia; MPI, OpenMPI, Linux/Bash; NumPy, SciPy, SUNDIALS/ODE, BLAS/LAPACK

Publications and Manuscripts

- **R. Dora**, R. Korol, V. Tiwari, R. Chourasiya, I. Franco, "Light–matter interactions beyond the electric dipole approximation in extended systems without multipole expansion," arXiv:2603.10271; *Physical Review B* (under review).

Manuscripts in Preparation

- **R. Dora**†, J. Fernandez†, I. Franco, "Light–field control of real and virtual charge carriers with general bichromatic waveforms." *Manuscript in preparation.*
- J. Fernandez †, **R. Dora**†, I. Franco, "Light–field control of real and virtual carriers in semiconductors, metallic and topological materials." *Manuscript in preparation.*
- **R. Dora**, I. Franco, "Light–field control in graphene nanojunctions beyond the electric-dipole approximation." *Manuscript in preparation.*

†Equal contribution.

Thesis

- **R. Dora**, "Polaron Transport in Optomechanical Arrays," BS–MS Thesis, Department of Physics, IISER Bhopal, 2022. Advisor: Prof. Sebastian Wüster.

Selected Presentations

Quantum Thermodynamics and Decoherence Symposium — June 2026
Poster: "Light–matter interactions beyond the dipole approximation" — *University of Rochester / University of Maryland*

University of Rochester CCQS Symposium — April 2026
Poster: "Light–matter interactions beyond the dipole approximation" — *University of Rochester*

Gordon Research Conference and Seminar – Quantum Control of Light and Matter — August 2025
Talk and Poster: "Light–matter interactions beyond the dipole approximation" — *Salve Regina University*

Center for Coherence and Quantum Science (CQS-12) — June 2025
Poster: "Light–matter interactions beyond the dipole approximation" — *University of Rochester*

Awards

Poster Session Award – 2nd Prize — June 2026
Quantum Thermodynamics and Decoherence Symposium — *University of Rochester / University of Maryland*

Science Academies' Summer Research Fellow — May 2019 – July 2019
Indian Academy of Sciences — *ARIES, Nainital*

Research Mentoring

Undergraduate Research Mentor — 2026 – Present
University of Rochester

- Mentor an undergraduate researcher studying electromagnetic fields in nanostructures using Python and MEEP/FDTD simulations, providing guidance on computational modeling, data analysis, debugging, and research planning.

Academic Service & Outreach

CHEMeliora Alumni Connections Subcommittee — 2026 – Present
Member, University of Rochester

- Support alumni engagement, graduate-career programming, and departmental career-development initiatives.

Rothschild Lectureship Committee — 2026 – Present
Graduate Student Representative, University of Rochester

- Collaborate with graduate-student representatives in identifying and selecting distinguished speakers for the annual Rothschild Lectureship.

CHEMeliora – Diversity, Equity and Inclusion Committee — 2024 – Present
Group Ambassador, University of Rochester

- Facilitate communication between the committee and research-group members and gather feedback on departmental initiatives.

International Students and Scholars Advisory Board — 2024 – 2025
Member, University of Rochester

- Served as a liaison between the International Services Office and the international-student community.

References

Prof. Ignacio Franco — ignacio.franco@rochester.edu
Department of Chemistry, University of Rochester

Prof. Roman Korol — Roman.Korol@USherbrooke.ca
Department of Chemistry, Sherbrooke University

Prof. David McCamant — david.mccamant@rochester.edu
Department of Chemistry, University of Rochester